MAX LORSIGNOL

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Max is a geomatics specialist with a background in GIS with experience in Photogrammetry, Ortho, LiDAR, and remote sensing. He recently completed a Master's in Geomatics for Environmental Management at UBC, where he focused on spatial analysis and environmental applications of geospatial data. His experience in aerial survey quality control and workflow processing has strengthened his skills in data accuracy, reproducibility, and practical problem-solving. With 4 years of experience with deploying remote sensing platforms for data collection, Max is motivated in using spatial data to better understand landscapes and urban systems and enjoys contributing to collaborative, multidisciplinary projects.

Education

Masters of Geomatics for Environmental Management – University of British Columbia Vancouver, 2025

BSc Earth and Environmental Science – University of British Columbia Okanagan, 2019

Employment History

Aeroquest Mapcon, Coquitlam, BC – Photogrammetry and LiDAR QC Specialist, (June 2025 – present)

AquaCoustic Remote Technologies Inc, Vancouver, BC – Operator & Field Technician, (2020 – 2024)

Expertise

- ArcGIS, QGIS, Global Mapper
- LiDAR, GPS, Imagery, Orthophoto, SONAR
- Technical reporting, Map making, Illustrating, Photoshop
- R, Python
- Microsoft Office (Excel, PowerPoint, SharePoint)

Select Project Experience

Remote Sensing & Geospatial Data Processing

- Aerial Survey QC – Processed and quality-controlled LiDAR, Photo, and GPS datasets for 10–30 concurrent projects. Automated reprojection, metadata parsing, and QA reporting to improve reproducibility and efficiency.
- Geomatic Surveying for Environmental Monitoring – Collected and processed SONAR, LiDAR, and CCTV data via ROVs and surface platforms. Integrated multi-sensor datasets into GIS workflows for hydrological infrastructure mapping.
- Forest Inventory & Structure from LiDAR – Used ArcGIS Pro and R to turn airborne LiDAR into decision-ready forest metrics by building simple models to map biomass/height/volume from CHMs and compared individual-tree detection approaches.

Environmental & Conservation Analysis

- OECM Candidate Identification in the Átl'ka7tsem / Howe Sound Biosphere – Developed a weighted suitability model integrating slope, forest age, watershed, and governance data to locate potential Other Effective area-based Conservation Measures. Published framework supporting Canada's 30×30 biodiversity goals.
- Marbled Murrelet Habitat Suitability Assessment – Analyzed historical critical habitat, canopy structure, and connectivity metrics using satellite imagery and forest data to evaluate conservation and restoration opportunities for at-risk bird species.
- Green-Space Equity Analysis for City of Vancouver – Combined census data with NDVI and park coverage in ArcGIS Pro/R to identify patterns of access to urban greenery and highlight areas for targeted investment.

Research & Publications

"Optimizing Canada's Conservation Strategy: OECM Candidate Identification in Átl'ka7tsem / Howe Sound."

- Published framework integrating spatial datasets and zoning/governance for OECM recognition
- Created a scalable, weighted suitability model to identify 260+ candidate parcels in Howe Sound Biosphere Region
- Provided a replicable methodology aligned with federal biodiversity targets (30x30)

Hobbies

Rock climbing, hiking, skiing, mountain biking, camping. Wild plant and mushroom foraging, plant identification. Wood working and metal fabrication. Reading and drawing. Volunteering for Stanley Park Ecology.